

IRARR Driver User Manual

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5-Sensor IR Array Line Detection System | Version 1.0

1. Overview

The **IRARR Driver** is a specialized software interface designed for an 8-bit AVR microcontroller (specifically ATmega32A) to interact with a 5-sensor infrared reflectance array. Unlike basic binary detection, this driver calculates a **weighted position**, providing high-resolution feedback on where a line is located relative to the robot's center.

2. How It Works

The driver operates on a "weighted average" principle. Each of the five sensors is assigned a fixed coordinate weight based on its physical location on the sensor bar.

Sensor Layout & Weights

Sensor	S0 (Far Left)	S1 (Near Left)	S2 (Center)	S3 (Near Right)	S4 (Far Right)
Weight	-2	-1	0	+1	+2

The Logic Loop

- Data Acquisition:** The driver reads digital signals from 5 pins on a shared PORT.
- Error Calculation:** Each reading (0 or 1) is multiplied by its corresponding weight.
- This sum is the `IRARR_error`.
Density Check: The driver counts how many sensors see the line
- (`IRARR_weightSum`).
Scaling: To avoid slow floating-point math, the result is multiplied by 100.

$$Position = (Error \times 100) / WeightSum$$

3. Technical Specifications

- **Microcontroller:** AVR (ATmega32A)
- **Port Configuration:** PORTC [Pins 0-4]
- **Return Type:** signed 16-bit integer (si16)
- **Resolution Range:** -100 (Full Left) to +100 (Full Right)
- **Special Status:** 1000 (Line Lost / No sensors active)

4. API Reference

void IRARR_init(void)

Initializes the hardware pins. It configures the designated pins on PORTC as digital inputs.

si16 IRARR_read(void)

Performs a sensor sweep and returns the calculated position of the line.

ui8 IRARR_ReadingsArr[5]

A public array containing the raw binary values (0 or 1) of the last successful reading. Useful for debugging individual sensor health.

5. Quick Start Example

```
#include "irarr_driver.h"

int main(void) {
    // Initialize GPIO and IR Array
    IRARR_init();

    while(1) {
        si16 linePos = IRARR_read();
```

```
    if (linePos == 1000) {  
        // ACTION: Stop motors or enter search mode  
    } else if (linePos < 0) {  
        // ACTION: Adjust steering LEFT  
    } else if (linePos > 0) {  
        // ACTION: Adjust steering RIGHT  
    } else {  
        // ACTION: Move STRAIGHT  
    }  
}  
}
```
